

APPLIED PHYSICS (BPHY-001) — MASTER NOTES

Complete University Standard Academic Courseware • Theory, Formal Derivations, Ray/Energy Schematics & Solved Proofs

Institution: Maharishi Markandeshwar (Deemed to be University), Mullana **Program:** B. Tech. 1st Year (Batch 2025 onwards)

Course Credits: 4.0 (L: 3, T: 1, P: 0) **Assessment Scheme:** Continuous: 40 Marks | End-Sem: 60 Marks | Total: 100 Marks

UNIT I: WAVE OPTICS & LASERS

WAVEFRONTS • INTERFERENCE • DIFFRACTION • POLARIZATION • QUANTUM LASER SYSTEMS

1. Principles of Wave Optics & Superposition

Huygens' Principle: Every point on an advancing primary wavefront acts as a secondary source of spherical wavelets that spread out in all directions with the speed of light in that medium. The forward envelope (tangential surface) to these secondary wavelets at any later instant constitutes the new primary wavefront.

Division of Wavefront

The primary wavefront incident from a point source is physically split into two distinct coherent sub-wavefronts using apertures, prisms, or mirrors.

- **Classical Examples:** Young's Double Slit Experiment (YDSE), Fresnel Biprism, Lloyd's Mirror.
- **Requirement:** Requires a narrow slit or point source of light.

Division of Amplitude

The amplitude (and intensity) of an incident light wave is split into two or more parts by partial reflection and partial refraction at one or more bounding interfaces.

- **Classical Examples:** Interference in Thin Dielectric Films, Wedge-shaped Films, **Newton's Rings**, Michelson Interferometer.
- **Requirement:** Broad/extended light sources can be used, producing localized fringes.

2. Interference in Thin Films & Newton's Rings (Complete Derivation)

Optical Path Difference in Thin Films: When a plane light wave of wavelength λ enters a thin transparent film of thickness t and refractive index μ at an angle of refraction r , the geometric path difference between rays reflected from upper and lower surfaces is $\Delta_{\text{geom}} = 2\mu t \cos r$.

Stokes' Law of Phase Reversal upon Reflection

Boundary Condition

When light reflects from the boundary of an optically denser medium (air-to-glass interface), an abrupt phase change of π radians occurs, corresponding to an additional path difference of $\pm \lambda / 2$. No phase jump occurs when reflecting from a rarer medium.

$$\text{Effective Optical Path Difference: } \Delta = 2\mu t \cos(r) - \lambda / 2 \quad (\text{for reflected light})$$

Mathematical Derivation: Radii of Newton's Rings (Reflected System)

A plano-convex lens of large radius of curvature R is placed with its curved surface in contact with an optically flat glass plate. An air film of variable thickness t is enclosed. At a radial distance r_n from the point of contact O :

From the geometric property of intersecting chords in a sphere of radius R : $r_n^2 = t(2R - t)$. Since film thickness $t \ll R$, t^2 is negligible:

$$2t = r_n^2 / R \quad \text{or} \quad t = r_n^2 / (2R) = D_n^2 / (8R) \quad [\text{where } D_n = 2r_n \text{ is ring diameter}]$$

For normal incidence ($\cos r = 1$) in an air film ($\mu = 1$), the net path difference is $\Delta = 2t - \lambda/2$.

Condition for Dark Rings (Destructive Interference):

$$\Delta = 2t - \lambda/2 = (2n - 1)\lambda/2 \Rightarrow 2t = n\lambda \quad (\text{where } n = 0, 1, 2, \dots)$$

$$D_n^2 / (4R) = n\lambda \Rightarrow D_n = 2\sqrt{n\lambda R}$$

$D_n \propto \sqrt{n}$: Diameters of dark rings are directly proportional to the square roots of consecutive natural numbers (0, 1, 2, 3...).

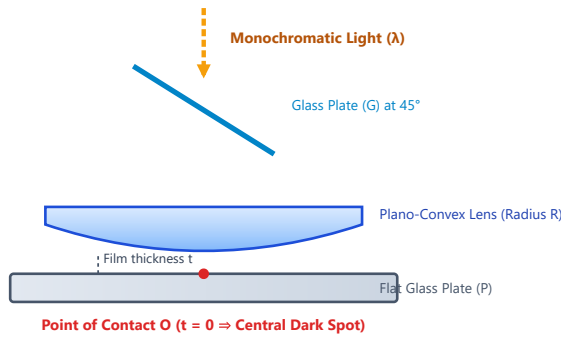
Condition for Bright Rings (Constructive Interference):

$$\Delta = 2t - \lambda/2 = n\lambda \Rightarrow 2t = (2n - 1)\lambda/2 \quad (\text{where } n = 1, 2, 3, \dots)$$

$$D_n^2 / (4R) = (2n - 1)\lambda/2 \Rightarrow D_n = \sqrt{2(2n - 1)\lambda R}$$

$D_n \propto \sqrt{(2n - 1)}$: Diameters of bright rings are proportional to square roots of consecutive odd numbers (1, 3, 5...).

Newton's Rings Experimental Setup (Air Wedge Geometry)



Reflected Circular Fringe Pattern

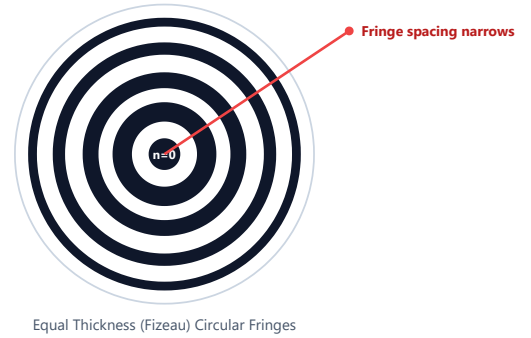


FIGURE 1.1: NEWTON'S RINGS EXPERIMENTAL APPARATUS & CONCENTRIC CIRCULAR INTERFERENCE PATTERN

Determination of Wavelength (λ)

Let D_{n+p} and D_n be diameters of the $(n+p)^{\text{th}}$ and n^{th} dark rings:

$$D_{n+p}^2 = 4(n+p)\lambda R \quad \text{and} \quad D_n^2 = 4n\lambda R$$

$$\lambda = [D_{n+p}^2 - D_n^2] / [4pR]$$

Taking differences eliminates error due to imperfect optical contact at point O.

Determination of Refractive Index (μ) of Liquid

When a liquid of refractive index μ is introduced between lens and plate, optical path difference becomes $2\mu t - \lambda/2$:

$$\mu = [D_{n+p}^2 - D_n^2]_{\text{air}} / [D_{n+p}^2 - D_n^2]_{\text{liquid}}$$

Since $\mu > 1$, rings contract inward in the presence of the liquid.

3. Diffraction: Fresnel vs. Fraunhofer & Single Slit Intensity Distribution

Parameter	Fresnel Diffraction	Fraunhofer Diffraction
Source & Screen Distance	Source and screen are at finite distances from the diffracting aperture/obstacle.	Source and screen are effectively at infinite distances from the aperture.
Incident Wavefront	Spherical or cylindrical wavefront.	Strictly plane wavefront .
Use of Lenses	No lenses required to produce or observe fringes.	Requires two convex lenses (collimating & focusing).
Mathematical Treatment	Complex; involves Fresnel half-period zones and integrals.	Simple; involves plane trigonometry, phasors, and Fourier transforms.
Fringe Geometry	Fringes are shadows of the obstacle; non-uniform curvature.	Fringes are exact spectral images of the illuminating slit.
Central Point Intensity	Can be bright or dark depending on zone clearance.	Always a broad, intense Central Principal Maximum .

Derivation of Fraunhofer Single Slit Diffraction Intensity

Consider a slit AB of width a illuminated normally by a monochromatic plane wave λ . The slit is divided into N equal elementary strips of width $dx = a/N$. The phase difference between wavelets from successive strips at angle θ is $d\phi = (2\pi/\lambda) dx \sin\theta$.

Using the polygon method for N phasors each of amplitude a_0 , the resultant amplitude R is:

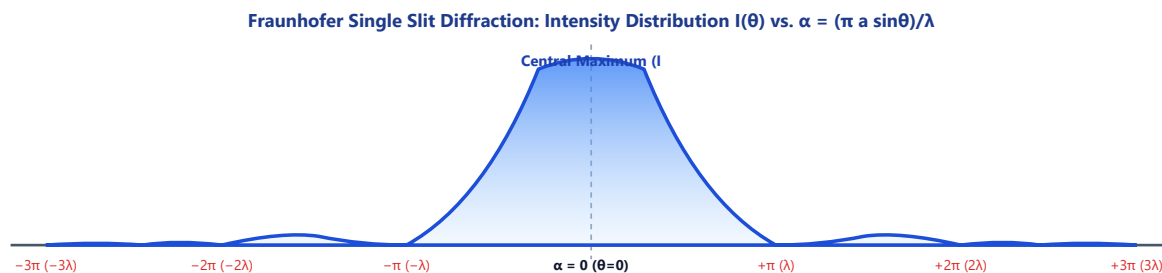
$$R = A_0 \cdot [\sin(\alpha) / \alpha] \quad \text{where} \quad \alpha = (\pi a \sin\theta) / \lambda \quad \text{and} \quad I(\theta) = I_0 [\sin(\alpha) / \alpha]^2$$

- **Central Maximum:** As $\theta \rightarrow 0$, $\alpha \rightarrow 0$, and $\lim(\alpha \rightarrow 0) [\sin(\alpha)/\alpha] = 1 \Rightarrow I = I_0$ (Maximum intensity).
- **Positions of Minima:** $I = 0$ when $\sin(\alpha) = 0$ but $\alpha \neq 0 \Rightarrow \alpha = \pm m\pi$ (where $m = 1, 2, 3, \dots$):

$$a \sin\theta = \pm m \lambda \quad (\text{Condition for } m^{\text{th}} \text{ Minimum})$$

- **Positions of Secondary Maxima:** $dI/d\alpha = 0 \Rightarrow \tan(\alpha) = \alpha$. Solved graphically: $\alpha \approx \pm 1.43\pi, \pm 2.46\pi, \pm 3.47\pi$.

- **Intensity Ratio of Maxima:** $I_0 : I_1 : I_2 : I_3 = 1 : (2/3\pi)^2 : (2/5\pi)^2 : (2/7\pi)^2 \approx 1 : 0.045 : 0.016 : 0.008$. The central peak contains over 95% of total energy!



I_0 1st Secondary Max (4.5%) 2nd (1.6%) Angular Width of Central Maximum: $2\theta = 2\lambda / a$ (Linear Width = $2f\lambda / a$)

FIGURE 1.2: FRAUNHOFER SINGLE SLIT DIFFRACTION: RELATIVE INTENSITY CURVE SHOWING RAPID SIDE-LOBE DECAY

4. Diffraction Grating & Resolving Power

A diffraction grating consists of N parallel, equidistant slits each of width a separated by opaque spaces of width b . The grating element is $d = (a + b)$. The number of lines per unit length is $N' = 1 / (a + b)$.

Grating Equation & Absent Spectra

Principal Maxima Condition: $(a + b) \sin\theta = n\lambda$ ($n = 0, 1, 2, \dots$)

Absent (Missing) Spectra: Occurs when the condition for a principal maximum coincides with the condition for a single-slit minimum ($a \sin\theta = m\lambda$):

$$n / m = (a + b) / a \quad \text{or} \quad n = m \cdot [(a + b) / a]$$

If opaque space equals slit width ($b = a$), then $n = 2m \Rightarrow$ Orders 2, 4, 6... are completely absent from the spectrum!

Rayleigh's Criterion & Resolving Power

Rayleigh Criterion: Two spectral lines are just resolved when the central principal maximum of one falls directly upon the first minimum of the other.

$$\text{Resolving Power: } R = \lambda / d\lambda = n \cdot N$$

where n is the diffraction order and N is the total number of illuminated grating lines.

5. Polarization of Light & Birefringence

Light is an electromagnetic transverse wave where the electric field vector $\mathbf{E}$ vibrates perpendicular to the direction of propagation. Unpolarized light consists of random vibration directions; linearly polarized light restricts vibration to a single plane.

Brewster's Law

When unpolarized light strikes a dielectric interface at polarizing angle i_p , reflected light is 100% plane-polarized perpendicular to plane of incidence. Reflected and refracted rays are orthogonal ($r + i_p = 90^\circ$):

$$\mu = \tan(i_p)$$

Malus's Law

When completely plane-polarized light of intensity I_0 passes through an analyzer rotated by angle θ relative to transmission axis of polarizer:

$$I = I_0 \cos^2(\theta)$$

When $\theta = 90^\circ$ (crossed nicols), transmitted intensity is zero.

Retardation Plates

Quarter-Wave Plate (QWP): Introduces path difference $\lambda/4$ (phase difference $\pi/2$) between O-ray and E-ray:

$$t = \lambda / [4(\mu_o - \mu_e)]$$

Half-Wave Plate (HWP): Introduces path difference $\lambda/2$ (π phase):

$$t = \lambda / [2(\mu_o - \mu_e)]$$

6. Lasers: Fundamentals, Einstein Coefficients & He-Ne Systems

LASER stands for *Light Amplification by Stimulated Emission of Radiation*. Lasers produce light with unprecedented: (i) **High Monochromaticity** ($\Delta\lambda \approx 10^{-6} \text{ \AA}$), (ii) **Directionality** (divergence $< 1 \text{ mrad}$), (iii) **Spatial & Temporal Coherence** (coherence length kilometers long), and (iv) **Extreme Intensity**.

Einstein's A and B Coefficients & Thermodynamic Derivation

Consider two atomic energy levels E_1 (ground) and E_2 (excited) with populations N_1 and N_2 in thermal equilibrium with radiation field of spectral energy density $u(\nu)$ at temperature T :

1. **Stimulated Absorption:** $R_{\text{abs}} = B_{12} N_1 u(\nu)$ [B_{12} : Einstein absorption coefficient]
2. **Spontaneous Emission:** $R_{\text{spont}} = A_{21} N_2$ [A_{21} : Einstein spontaneous transition probability]
3. **Stimulated Emission:** $R_{\text{stim}} = B_{21} N_2 u(\nu)$ [B_{21} : Einstein stimulated emission coefficient]

At thermal equilibrium: Rate of Upward Transitions = Rate of Downward Transitions:

$$B_{12} N_1 u(\nu) = A_{21} N_2 + B_{21} N_2 u(\nu) \Rightarrow u(\nu) = A_{21} / [(B_{12} N_1 / N_2) - B_{21}]$$

From Maxwell-Boltzmann statistics: $N_1 / N_2 = \exp[(E_2 - E_1) / k_B T] = \exp[h\nu / k_B T]$. Substituting into equilibrium equation:

$$u(\nu) = [A_{21} / B_{21}] \cdot [1 / \{ (B_{12}/B_{21}) e^{h\nu/k_B T} - 1 \}]$$

Comparing with **Planck's Radiation Law:** $u(\nu) = [8\pi h\nu^3 / c^3] \cdot [1 / (e^{h\nu/k_B T} - 1)]$:

Identity 1: $B_{12} = B_{21}$ (Probabilities are equal)

Identity 2: $A_{21} / B_{21} = 8\pi h \nu^3 / c^3 \propto \nu^3$

Physical Consequence: The ratio of spontaneous to stimulated emission grows as ν^3 , explaining why X-ray lasers are vastly more difficult to pump than optical or microwave (MASER) devices.

Population Inversion & Metastable State

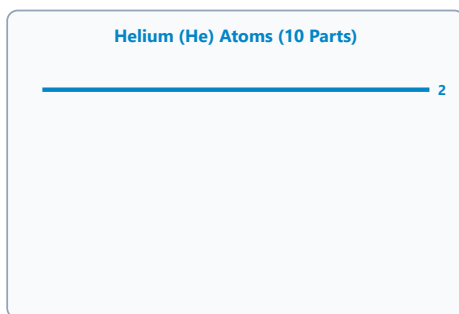
Under normal thermal equilibrium, $N_1 > N_2$. To achieve net optical amplification, one must establish **Population Inversion:** $N_2 > N_1$.

This requires a **Metastable State:** an excited atomic level with an exceptionally long lifetime ($\tau \approx 10^{-3}$ s compared to normal 10^{-8} s), allowing atoms to accumulate.

Pumping Mechanisms

- **Optical Pumping:** Intense xenon flash lamps (used in Ruby laser).
- **Electric Discharge:** High voltage DC collision ionization (used in He-Ne gas laser).
- **Inelastic Atom-Atom Collision:** Resonant energy transfer between distinct atomic species (He to Ne).
- **Chemical Pumping:** Exothermic chemical bond dissociation.

Four-Level Helium-Neon (He-Ne) Gas Laser: Resonant Energy Transfer & Transitions



1^1S (20.61 eV) 2^3S (19.82 eV) 1^1S Ground (0 eV) e- Collision Pumping Neon (Ne) Atoms (1 Part - Active Medium) $3s$ (20.66 eV) $2s$ (19.78 eV) $2p$ Level $1s$ Metastable Ground (0 eV) Resonant Energy Transfer ($\Delta E \approx 0.05$ eV) LASER: $\lambda = 632.8$ nm (Red) Fast Radiative (600 nm) Collisions with Tube Wall

FIGURE 1.3: ENERGY LEVEL DIAGRAM OF HE-NE FOUR-LEVEL LASER SHOWING RESONANT COLLISION & LASER EMISSION

UNIT II: ELECTROSTATICS, MAGNETOSTATICS & ELECTROMAGNETIC WAVES

MAXWELL'S EQUATIONS • CONTINUITY • WAVE EQUATION • SKIN DEPTH • POYNTING VECTOR

1. Foundations of Electrostatics & Magnetostatics

Electrostatics: Div, Curl & Potential

Gauss's Law (Div E): The total outward electric flux through any closed Gaussian surface equals the enclosed charge

Magnetostatics: Div, Curl & Vector Potential

Gauss's Law for Magnetism: Magnetic monopoles do not exist; magnetic flux lines form closed continuous loops:

divided by ϵ_0 :

$$\nabla \cdot \mathbf{E} = \rho / \epsilon_0 \quad (\text{Differential}) \Leftrightarrow \oint \mathbf{E} \cdot d\mathbf{A} = Q_{\text{enc}} / \epsilon_0$$

Conservative Nature (Curl $\mathbf{E}$): Electrostatic field is conservative, meaning line integral around any closed loop is zero:

$$\nabla \times \mathbf{E} = 0 \Rightarrow \mathbf{E} = -\nabla V$$

Substituting into Gauss's Law yields **Poisson's Equation**:

$$\nabla^2 V = -\rho / \epsilon_0 \quad [\text{In charge-free region: } \nabla^2 V = 0 \text{ (Laplace Eq.)}]$$

$$\nabla \cdot \mathbf{B} = 0 \quad (\text{Differential}) \Leftrightarrow \oint \mathbf{B} \cdot d\mathbf{A} = 0$$

Ampere's Circuital Law (Static):

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} \quad (\text{Differential}) \Leftrightarrow \oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enc}}$$

Since $\nabla \cdot \mathbf{B} = 0$, we define **Magnetic Vector Potential $\mathbf{A}$** :

$$\mathbf{B} = \nabla \times \mathbf{A} \quad \text{with Coulomb Gauge: } \nabla \cdot \mathbf{A} = 0$$

Yielding vector Poisson equation: $\nabla^2 \mathbf{A} = -\mu_0 \mathbf{J}$.

2. Maxwell's Modification: Displacement Current & Fundamental Equations

Derivation of Displacement Current from Charge Continuity

Equation of Continuity: Conservation of electric charge requires that net current flowing out of a closed volume equals rate of charge decrease inside: $\nabla \cdot \mathbf{J} + \partial \rho / \partial t = 0$.

Taking the divergence of static Ampere's Law: $\nabla \cdot (\nabla \times \mathbf{B}) = \mu_0 (\nabla \cdot \mathbf{J})$. Vector calculus dictates divergence of any curl is identically zero: $\nabla \cdot (\nabla \times \mathbf{B}) \equiv 0$. This would demand $\nabla \cdot \mathbf{J} = 0$, which violates continuity for time-varying fields ($\partial \rho / \partial t \neq 0$)!

Maxwell resolved this contradiction by replacing ρ using Gauss's Law ($\rho = \epsilon_0 \nabla \cdot \mathbf{E}$):

$$\nabla \cdot \mathbf{J} + \partial(\epsilon_0 \nabla \cdot \mathbf{E}) / \partial t = 0 \Rightarrow \nabla \cdot [\mathbf{J} + \epsilon_0 \partial \mathbf{E} / \partial t] = 0$$

Thus, total current density consists of conduction current density $\mathbf{J}$ plus **Displacement Current Density $\mathbf{J}_d$** :

$$\mathbf{J}_d = \partial \mathbf{D} / \partial t = \epsilon_0 \partial \mathbf{E} / \partial t$$

Law	Differential Form	Integral Form	Physical Significance
Gauss's Law (Electrostatics)	$\nabla \cdot \mathbf{E} = \rho / \epsilon_0$	$\oint \mathbf{E} \cdot d\mathbf{A} = Q_{\text{enc}} / \epsilon_0$	Electric charges act as sources and sinks of electric fields.
Gauss's Law (Magnetism)	$\nabla \cdot \mathbf{B} = 0$	$\oint \mathbf{B} \cdot d\mathbf{A} = 0$	Isolated magnetic poles (monopoles) do not exist in nature.
Faraday's Law of Induction	$\nabla \times \mathbf{E} = -\partial \mathbf{B} / \partial t$	$\oint \mathbf{E} \cdot d\mathbf{l} = -d/dt \int \mathbf{B} \cdot d\mathbf{A}$	A time-varying magnetic flux generates a circulating electric field.
Ampere-Maxwell Law	$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \partial \mathbf{E} / \partial t$	$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_c + \mu_0 \epsilon_0 d/dt \int \mathbf{E} \cdot d\mathbf{A}$	Magnetic fields are generated by conduction currents and changing electric fields.

3. Electromagnetic Wave Propagation & Velocity in Vacuum

Derivation of EM Wave Equation in Free Space

In source-free vacuum ($\rho = 0, \mathbf{J} = 0$):

$$(1) \nabla \cdot \mathbf{E} = 0 \quad (2) \nabla \cdot \mathbf{B} = 0 \quad (3) \nabla \times \mathbf{E} = -\partial \mathbf{B} / \partial t \quad (4) \nabla \times \mathbf{B} = \mu_0 \epsilon_0 \partial \mathbf{E} / \partial t$$

Take curl of Faraday's Law (3): $\nabla \times (\nabla \times \mathbf{E}) = -\partial(\nabla \times \mathbf{B}) / \partial t$

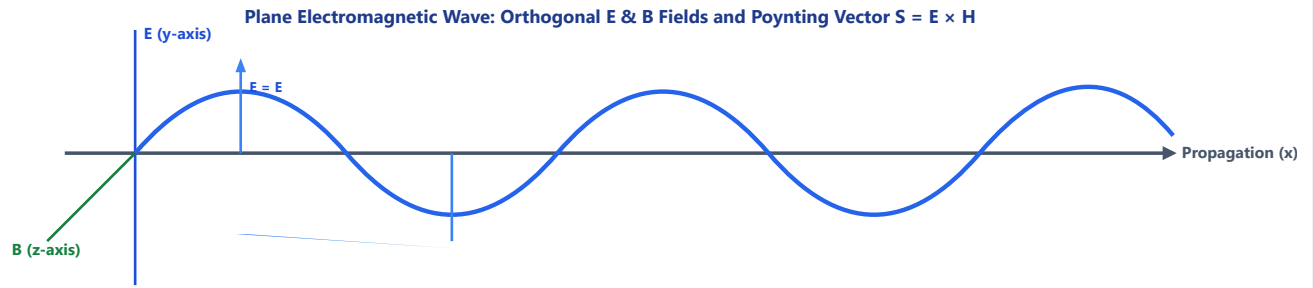
Using vector identity $\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E}$, and substituting (1) and (4):

$$0 - \nabla^2 \mathbf{E} = -\partial(\mu_0 \epsilon_0 \partial \mathbf{E} / \partial t) / \partial t \Rightarrow \nabla^2 \mathbf{E} = \mu_0 \epsilon_0 \partial^2 \mathbf{E} / \partial t^2$$

Comparing with the 3D classical wave equation $\nabla^2 \mathbf{E} = (1/v^2) \partial^2 \mathbf{E} / \partial t^2$ proves EM radiation propagates with speed:

$$c = 1 / \sqrt{(\mu_0 \epsilon_0)} = 1 / \sqrt{[(4\pi \times 10^{-7}) \times (8.854 \times 10^{-12})]} \approx 2.998 \times 10^8 \text{ m/s}$$

Similarly for magnetic field: $\nabla^2 \mathbf{B} = \mu_0 \epsilon_0 \partial^2 \mathbf{B} / \partial t^2$.



$$E = E_0 \sin(kx - \omega t) \quad B = B_0 \sin(kx - \omega t) \quad \text{Transverse Nature: } \mathbf{k} \cdot \mathbf{E} = 0, \mathbf{k} \cdot \mathbf{B} = 0 \quad E_0 / B_0 = c \quad | \quad Z_0 = \sqrt{(\mu_0 / \epsilon_0)} \approx 377 \Omega$$

FIGURE 2.1: SPATIAL TOPOLOGY OF TRANSVERSE EM WAVE IN FREE SPACE SHOWING ORTHOGONAL FIELD VECTORS

4. Wave Propagation in Conducting Media & Skin Depth

Derivation of Attenuation Constant & Skin Depth (δ)

In a conducting medium having conductivity σ , permittivity ϵ , and permeability μ , Ohm's law gives conduction current $\mathbf{J} = \sigma \mathbf{E}$. The wave equation becomes:

$$\nabla^2 \mathbf{E} = \mu \sigma \partial \mathbf{E} / \partial t + \mu \epsilon \partial^2 \mathbf{E} / \partial t^2$$

For a harmonic plane wave $\mathbf{E}(z, t) = E_0 e^{i(\omega t - \gamma z)}$, the propagation constant is $\gamma^2 = -i\omega\mu\sigma + \omega^2\mu\epsilon$. We define $\gamma = \alpha + i\beta$:

For a **good conductor** ($\sigma \gg \omega\epsilon$), the conduction term dominates: $\gamma^2 \approx -i\omega\mu\sigma = \omega\mu\sigma e^{-i\pi/2}$. Taking square root:

$$\gamma = \alpha + i\beta = \sqrt{(\omega\mu\sigma)} \cdot e^{-i\pi/4} = \sqrt{(\omega\mu\sigma)} [\cos(\pi/4) - i \sin(\pi/4)] = \sqrt{[\omega\mu\sigma / 2]} - i \sqrt{[\omega\mu\sigma / 2]}$$

Thus, the attenuation constant is: $\alpha = \sqrt{[\omega\mu\sigma / 2]} = \sqrt{[\pi f \mu \sigma]}$.

The spatial wave amplitude decays according to $\mathbf{E}(z) = \mathbf{E}_0 e^{-\alpha z}$.

$$\text{Skin Depth: } \delta = 1 / \alpha = \sqrt{2 / (\omega \mu \sigma)} = 1 / \sqrt{\pi f \mu \sigma}$$

Physical Definition of Skin Depth: Skin depth (δ) is the penetration depth into a conducting material at which the amplitude of the electromagnetic field attenuates to $1/e$ ($\approx 36.8\%$) of its surface value.

Engineering Applications: Explains why radio signals cannot penetrate deep seawater, why RF coaxial cables use silver plating (current flows only in outer microscopic skin), and electromagnetic shielding of sensitive avionics.

5. Energy Flow: Poynting Vector & Poynting Theorem

Mathematical Derivation of Poynting's Theorem (Work-Energy Balance)

Take dot product of Faraday's Law with $\mathbf{H}$ and Ampere's Law with $\mathbf{E}$:

$$(1) \mathbf{H} \cdot (\nabla \times \mathbf{E}) = -\mu \mathbf{H} \cdot \partial \mathbf{H} / \partial t = -\partial / \partial t [1/2 \mu H^2]$$

$$(2) \mathbf{E} \cdot (\nabla \times \mathbf{H}) = \mathbf{E} \cdot \mathbf{J} + \epsilon \mathbf{E} \cdot \partial \mathbf{E} / \partial t = \mathbf{J} \cdot \mathbf{E} + \partial / \partial t [1/2 \epsilon E^2]$$

Subtracting (2) from (1) and utilizing vector identity $\nabla \cdot (\mathbf{E} \times \mathbf{H}) = \mathbf{H} \cdot (\nabla \times \mathbf{E}) - \mathbf{E} \cdot (\nabla \times \mathbf{H})$:

$$-\nabla \cdot (\mathbf{E} \times \mathbf{H}) = \partial / \partial t [1/2 \epsilon E^2 + 1/2 \mu H^2] + \mathbf{J} \cdot \mathbf{E}$$

Integrating over an arbitrary volume V enclosed by surface S and applying Gauss's Divergence Theorem:

$$- \oint_S (\mathbf{E} \times \mathbf{H}) \cdot d\mathbf{A} = \partial / \partial t \int_V [1/2 \epsilon E^2 + 1/2 \mu H^2] dV + \int_V (\mathbf{J} \cdot \mathbf{E}) dV$$

Left Side: $\oint \mathbf{S} \cdot d\mathbf{A}$

First Term: $\partial u_{EM} / \partial t$

Second Term: $\int \mathbf{J} \cdot \mathbf{E} dV$

Net rate of electromagnetic energy radiating outwards across boundary surface S .

$$S = E \times H \text{ (W/m}^2\text{)}$$

Time rate of decrease of total energy stored in electric and magnetic fields within V .

Rate of work done on charges (Ohmic Joule heating dissipation rate σE^2).

UNIT III: WAVE NATURE OF PARTICLES & SCHRÖDINGER EQUATION

DE BROGLIE • PHASE & GROUP VELOCITY • UNCERTAINTY • WAVEFUNCTION • 1D BOX

1. de Broglie Hypothesis & Wave-Particle Duality

In 1924, Louis de Broglie hypothesized that nature is symmetric: if electromagnetic radiation exhibits both wave and particle characteristics, matter (electrons, protons) must also exhibit wave properties when in motion.

$$\text{de Broglie Wavelength: } \lambda = h / p = h / (m v) = h / \sqrt{2 m E_k}$$

Wavelength for Accelerated Electron

An electron of charge e accelerated through potential difference V volts gains kinetic energy $E_k = eV$:

$$\lambda = h / \sqrt{2 m e V} = 1.227 / \sqrt{V} \text{ nm} = 12.27 / \sqrt{V} \text{ \AA}$$

For $V = 100 \text{ V}$, $\lambda = 1.227 \text{ \AA}$, perfectly comparable to crystal interplanar spacing (Davisson-Germer confirmation).

Phase Velocity vs. Group Velocity

Phase Velocity (v_p): Speed of individual wave crests: $v_p = \omega / k$.

Group Velocity (v_g): Speed of envelope packet conveying energy: $v_g = d\omega / dk$.

$$\text{Relation: } v_g = v_p - \lambda (dv_p/d\lambda) \quad | \quad v_p \cdot v_g = c^2 \text{ (Relativistic)}$$

Crucial Proof: de Broglie wave packet group velocity equals classical particle velocity: $v_g = v_{\text{particle}}$.

2. Heisenberg's Uncertainty Principle & Non-Existence of Electron in Nucleus

Heisenberg's Uncertainty Principle

Fundamental Postulate

It is fundamentally impossible to measure simultaneously both the position and canonical momentum of a particle with unlimited precision:

$$\Delta x \cdot \Delta p_x \geq \hbar / 2 \quad (\text{where } \hbar = h / 2\pi) \quad | \quad \Delta E \cdot \Delta t \geq \hbar / 2 \quad | \quad \Delta L \cdot \Delta \theta \geq \hbar / 2$$

Exemplary Application: Proof of Non-Existence of Free Electron inside the Nucleus

The nuclear diameter is typically $2R \approx 10^{-14} \text{ m}$. If an electron exists inside the nucleus, uncertainty in its position cannot exceed nuclear dimension: $\Delta x \approx 10^{-14} \text{ m}$.

By Heisenberg's uncertainty principle:

$$\Delta p \geq \hbar / (2 \Delta x) = (1.054 \times 10^{-34}) / [2 \times 10^{-14}] \approx 5.27 \times 10^{-21} \text{ kg}\cdot\text{m/s}$$

Using relativistic energy-momentum relation (since $p \gg m_0 c$): $E \approx p \cdot c$:

$$E \approx (5.27 \times 10^{-21}) \times (3 \times 10^8) = 1.58 \times 10^{-12} \text{ J} = [1.58 \times 10^{-12} / (1.6 \times 10^{-13})] \approx 9.87 \text{ MeV to } 20 \text{ MeV}$$

Conclusion: If an electron were trapped inside the nucleus, it must possess at least 10–20 MeV of kinetic energy. However, electrons emitted during nuclear β -decay have maximum energies of only 2 to 3 MeV. Hence, **electrons cannot exist as constituent particles inside atomic nuclei** (they are created at the instant of β -decay).

3. The Wave Function ψ , Born Interpretation & Schrödinger Equations

Physical Conditions for Well-Behaved ψ

Expectation Values of Observables

Max Born Postulate: Probability of finding a particle in volume dV is $P = |\psi|^2 dV = \psi^* \psi dV$.

- ψ must be **single-valued** at all points in space.
- ψ and its first spatial derivative $\partial\psi/\partial x$ must be **continuous** everywhere.
- ψ must be **square-integrable (normalizable)**:

$$\int_{-\infty}^{+\infty} |\psi|^2 dV = 1$$

The statistical expectation value of any quantum dynamical operator $\hat{A}$ is given by:

$$\langle A \rangle = \int \psi^* \hat{A} \psi dV / \int \psi^* \psi dV$$

- Position: $\langle x \rangle = \int \psi^* x \psi dx$
- Momentum: $\langle p_x \rangle = \int \psi^* [-i\hbar \partial/\partial x] \psi dx$
- Energy: $\langle E \rangle = \int \psi^* [i\hbar \partial/\partial t] \psi dx$

Derivation of Time-Dependent & Time-Independent Schrödinger Equations

A 1D de Broglie wave propagating along +x direction is represented as: $\psi(x, t) = A e^{i(kx - \omega t)} = A e^{i(px - Et)/\hbar}$.

1. Differentiating with respect to time t : $\partial\psi/\partial t = -(iE/\hbar) \psi \Rightarrow E \psi = i\hbar \partial\psi/\partial t$ [Energy Operator: $\hat{E} = i\hbar \partial/\partial t$].
2. Differentiating twice with respect to x : $\partial^2\psi/\partial x^2 = -(p^2/\hbar^2) \psi \Rightarrow p^2 \psi = -\hbar^2 \partial^2\psi/\partial x^2$ [Momentum: $\hat{p} = -i\hbar \partial/\partial x$].

Total classical energy: $E = p^2/(2m) + V(x, t)$. Multiplying by wave function ψ :

$$i\hbar \partial\psi/\partial t = - [\hbar^2 / (2m)] \partial^2\psi/\partial x^2 + V(x, t) \psi \quad \Leftrightarrow \quad i\hbar \partial\psi/\partial t = \hat{H} \psi \quad [1D \text{ TDSE}]$$

Using method of separation of variables: $\psi(x, t) = \phi(x) \cdot e^{-iEt/\hbar}$ yields the **Time-Independent Schrödinger Equation (TISE)**:

$$d^2\phi / dx^2 + [2m / \hbar^2] [E - V(x)] \phi = 0 \quad [1D \text{ TISE}]$$

4. Application: Particle in a 1D Infinite Potential Well (Rigid Box)

Rigorous Solution for Eigenfunctions and Energy Eigenvalues

Potential profile: $V(x) = 0$ for $0 \leq x \leq L$; and $V(x) = \infty$ for $x < 0$ and $x > L$.

Inside well ($V=0$), TISE is: $d^2\psi/dx^2 + k^2 \psi = 0$ where $k^2 = 2mE/\hbar^2$.

General solution: $\psi(x) = A \sin(kx) + B \cos(kx)$.

- **Boundary Condition 1:** At $x = 0$, $\psi(0) = 0 \Rightarrow A \sin(0) + B \cos(0) = 0 \Rightarrow B = 0$.
- **Boundary Condition 2:** At $x = L$, $\psi(L) = 0 \Rightarrow A \sin(kL) = 0$. Since $A \neq 0$ (trivial solution), $\sin(kL) = 0 \Rightarrow kL = n\pi$ ($n = 1, 2, 3, \dots$).

$$\text{Wavevector: } k_n = n\pi / L \quad \Rightarrow \quad E_n = \hbar^2 k_n^2 / (2m) = n^2 \pi^2 \hbar^2 / (2 m L^2) = n^2 h^2 / (8 m L^2)$$

Normalization: $\int_0^L |\psi_n(x)|^2 dx = 1 \Rightarrow A^2 \int_0^L \sin^2(n\pi x/L) dx = A^2 (L/2) = 1 \Rightarrow A = \sqrt{2/L}$.

$$\text{Normalized Eigenfunctions: } \psi_n(x) = \sqrt{2/L} \cdot \sin(n \pi x / L) \quad \text{for } n = 1, 2, 3, \dots$$

Zero-Point Energy ($n = 1$): $E_1 = h^2 / (8mL^2) \neq 0$. A particle in a box cannot have zero energy; if $E=0$, momentum $p=0$ ($\Delta p=0$), which violates uncertainty principle ($\Delta x \leq L \Rightarrow \Delta p > 0$).

$\psi_n(x)$ & Probability Densities $|\psi_n(x)|^2$ $x = 0$ $x = L$ $V(x) = 0$ inside box (Width L) $V = \infty$ $V = \infty$ $n=1$: $E_1 = h^2/(8mL^2)$ $n=2$: $E_2 = 4 E_1$ 1 Node $n=3$: $E_3 = 9 E_1$ 2 Nodes $\psi_n(x)$ Wavefunction $|\psi_n(x)|^2$ Prob. Density

FIGURE 3.1: ENERGY EIGENVALUES, SPATIAL MODES $\psi_N(x)$, AND PROBABILITY DENSITIES $|\psi_N(x)|^2$ FOR 1D INFINITE WELL

5. Finite Potential Well, Quantum Tunneling & Harmonic Oscillator

Quantum Tunneling (Barrier Penetration)

When a particle with energy E encounters a finite rectangular potential barrier of height $V_0 > E$ and width a , classical mechanics predicts 100% reflection.

Quantum mechanics predicts wave penetration inside barrier: $\psi_{II}(x) \sim e^{-\alpha x}$, resulting in a non-zero **Transmission Coefficient** T :

$$T \approx e^{-2\alpha a} \quad \text{where} \quad \alpha = \sqrt{2m(V_0 - E)} / \hbar$$

Physical Applications: α -decay from radioactive nuclei, Scanning Tunneling Microscope (STM), Tunnel Diodes.

1D Quantum Harmonic Oscillator

Potential energy is parabolic: $V(x) = 1/2 m \omega^2 x^2$.

Solving TISE via Hermite polynomials yields quantized energy eigenvalues:

$$E_n = (n + 1/2) \hbar \omega \quad (n = 0, 1, 2, 3, \dots)$$

- **Equispaced Levels:** $\Delta E = E_{n+1} - E_n = \hbar \omega = h\nu$.
- **Zero-Point Energy:** $E_0 = 1/2 \hbar \omega \neq 0$ (residual vibration at absolute zero).

UNIT IV: CRYSTAL STRUCTURES & FREE ELECTRON THEORY OF METALS

BRAVAIS LATTICES • SC/BCC/FCC DERIVATIONS • DEFECTS • SOMMERFELD QUANTUM THEORY • RICHARDSON EQ.

1. Crystallographic Fundamentals & Bravais Systems

Space Lattice: An infinite periodic mathematical array of points in 3D space where every point has identical surroundings. **Basis (Motif):** An atom or group of atoms associated with each lattice point.

$$\text{Crystal Structure} = \text{Space Lattice} + \text{Basis}$$

Unit Cell: The smallest geometric building block that, when repeated by regular translation along crystallographic axes (a , b , c), reproduces the entire macroscopic crystal. Primitive cells contain exactly 1 effective lattice point; non-primitive cells contain more than 1.

2. Derivations of Cubic Crystal Geometries (SC, BCC, FCC)

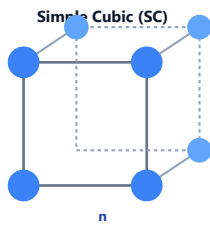
Step-by-Step Derivation of Coordination Number, Atomic Radius & APF

Atomic Packing Factor (APF): Ratio of volume occupied by atoms in unit cell to total unit cell volume:

$$\text{APF} = [n_{\text{eff}} \times (4/3 \pi r^3)] / a^3 \quad [\text{where } a \text{ is lattice parameter, } r \text{ is atomic radius}]$$

Property / Parameter	Simple Cubic (SC)	Body-Centered Cubic (BCC)	Face-Centered Cubic (FCC)
Corner Atom Share	$8 \times (1/8) = 1$	$8 \times (1/8) = 1$	$8 \times (1/8) = 1$
Center Atom Share	0	1 (at body center)	$6 \times (1/2) = 3$ (at face centers)
Effective Atoms (n_{eff})	1	2	4
Coordination Number (CN)	6 (6 nearest neighbors)	8 (8 corner neighbors)	12 (12 nearest neighbors)
Touching Geometry	Atoms touch along cube edge: $2r = a$	Atoms touch along body diagonal: $4r = \sqrt{3} a$	Atoms touch along face diagonal: $4r = \sqrt{2} a$
Atomic Radius (r)	$r = a / 2$	$r = (\sqrt{3} / 4) a$	$r = (\sqrt{2} / 4) a = a / (2\sqrt{2})$
Nearest Neighbor Dist. (d)	$d = 2r = a$	$d = 2r = (\sqrt{3} / 2) a$	$d = 2r = a / \sqrt{2}$
APF Calculation & %	$\text{APF} = [1 \times 4/3\pi(a/2)^3] / a^3$ $= \pi / 6 \approx \mathbf{52.4\%}$	$\text{APF} = [2 \times 4/3\pi(\sqrt{3} a/4)^3] / a^3$ $= \sqrt{3} \pi / 8 \approx \mathbf{68.0\%}$	$\text{APF} = [4 \times 4/3\pi(a/2\sqrt{2})^3] / a^3$ $= \sqrt{2} \pi / 6 \approx \mathbf{74.0\%}$
Representative Elements	Polonium (α -Po)	Fe (α -iron), Cr, W, Na	Cu, Al, Au, Ag, γ -Fe

Cubic Crystal Unit Cells: Sphere Packing, Touching Geometry & Lattice Parameters



$n_{\text{eff}} = 1$ | APF = 52% Body-Centered Cubic (BCC) $n_{\text{eff}} = 2$ | APF = 68% Face-Centered Cubic (FCC) $n_{\text{eff}} = 4$ | APF = 74% (Close-Packed)

FIGURE 4.1: UNIT CELL GEOMETRIES FOR SC, BCC, AND FCC LATTICES WITH COORDINATE ATOMS AND APF VALUES

3. Crystal Defects: Schottky vs. Frenkel Defects (12-Point Analysis)

Comparative Feature	Schottky Defect	Frenkel Defect
Fundamental Nature	Pair Vacancy Defect: Equal numbers of cations and anions are missing from their normal lattice sites.	Dislocation Defect: An ion (usually smaller cation) is displaced from normal lattice site to an interstitial site .
Stoichiometry & Charge	Crystal preserves strict electrical neutrality and stoichiometry.	Crystal preserves strict electrical neutrality and stoichiometry.
Size of Ions Involved	Favored when cation and anion are of nearly equal size (Radius ratio $r_+/r_- \approx 1$).	Favored when there is a large size difference between cation and anion ($r_+ \ll r_-$).
Coordination Number	Common in crystals with high coordination numbers (CN = 6 or 8).	Common in crystals with low coordination numbers (CN = 4).
Effect on Mass & Density	Mass decreases while volume remains constant $\Rightarrow$ Overall density decreases significantly.	No ions leave the crystal $\Rightarrow$ Density remains unchanged (constant).
Dielectric Constant (ϵ)	Negligible effect on dielectric constant.	Dielectric constant increases because similar charges are brought close together at interstitials.
Lattice Strains	Relatively small localized lattice relaxations.	Causes substantial lattice distortion and internal strain.
Equilibrium Defect Concentration	$n_s \approx N \cdot \exp[-E_s / (2 k_B T)]$ (E_s = energy required to form Schottky pair)	$n_f \approx \sqrt{N \cdot N_i} \cdot \exp[-E_f / (2 k_B T)]$ (N_i = number of interstitial positions)
Electrical Conductivity	Conductivity increases via vacancy hopping mechanism.	Conductivity increases via interstitial migration mechanism.
Entropy Effect	Creates configurational entropy through pair formation.	Creates configurational entropy through vacancy-interstitial pairing.
Prototypical Examples	NaCl, KCl, KBr, CsCl	AgCl, AgI, ZnS (Zn^{2+} and Ag^+ are small)
Special Case (Both Defects)	Silver Bromide (AgBr) exhibits <i>both</i> Schottky and Frenkel defects depending on thermal history.	

4. Free Electron Theory of Metals: Classical vs. Quantum Sommerfeld

Classical Theory (Drude-Lorentz) & Failures

Assumptions: Free valence electrons behave like an ideal monoatomic gas obeying Maxwell-Boltzmann classical statistics. Move randomly colliding with ion cores with relaxation time τ .

- Electrical Conductivity: $\sigma = n e^2 \tau / m$
- Wiedemann-Franz Law: $K / (\sigma T) = 3/2 (k_B/e)^2 = \text{Lorentz Number}$

Severe Failures of Classical Model:

- Electronic Specific Heat Anomaly:** Classical theory predicts electronic heat capacity $C_{v,el} = 3/2 R$. Observed value is $\sim 0.01 R$ at 300 K. Classical theory overestimates by a factor of 100!
- Paramagnetic Susceptibility:** Classical theory predicts $\chi \propto 1/T$ (Curie's Law). Experimentally, metal conduction electron paramagnetism is temperature-independent (Pauli paramagnetism).

Quantum Theory (Sommerfeld Postulates)

Assumptions: Conduction electrons are confined in a 3D potential box with constant potential $V=0$ inside. Obey **Pauli Exclusion Principle** and **Fermi-Dirac Statistics**.

Fermi-Dirac Distribution Function:

$$f(E) = 1 / [e^{(E - E_F)/(k_B T)} + 1]$$

- At $T = 0$ K:**
 - For $E < E_F$: $e^{-\infty} = 0 \Rightarrow f(E) = 1$ (All levels completely filled).
 - For $E > E_F$: $e^{+\infty} = \infty \Rightarrow f(E) = 0$ (All levels completely vacant).
- At $T > 0$ K:** Thermal energy $k_B T$ smears occupancy only near E_F . At $E = E_F$: $f(E_F) = 1 / (1 + 1) = 0.5$ (50% probability).

Derivation of Fermi Energy E_F at 0 K & Density of States

In a 3D potential cube of volume $V = L^3$, the number of quantum states per unit energy interval is given by the **Density of States** $g(E)$ (accounting for electron spin degeneracy 2):

$$g(E) dE = [V / (2\pi^2)] \cdot [2m / \hbar^2]^{3/2} \cdot E^{1/2} dE$$

At $T = 0$ K, all states up to $E_F(0)$ are completely filled ($f(E)=1$). Total number of conduction electrons N is:

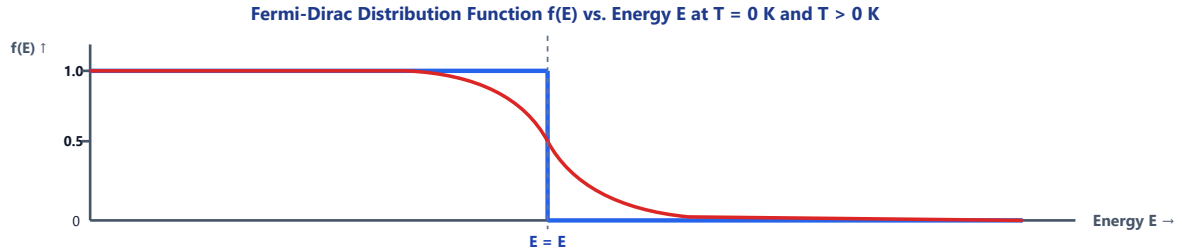
$$N = \int_0^{E_F(0)} g(E) dE = \left[V / (2\pi^2) \right] \left[2m / \hbar^2 \right]^{3/2} \int_0^{E_F(0)} E^{1/2} dE = \left[V / (3\pi^2) \right] \left[2m / \hbar^2 \right]^{3/2} [E_F(0)]^{3/2}$$

Solving for Fermi energy in terms of electron concentration $n = N/V$:

$$\text{Fermi Energy at } 0 \text{ K: } E_F(0) = \left[\hbar^2 / (2m) \right] \cdot \left[3\pi^2 n \right]^{2/3}$$

Fermi Velocity: $v_F = \sqrt{2 E_F / m} \approx 10^6$ m/s

Fermi Temperature: $T_F = E_F / k_B \approx 10^4$ to 10^5 K



f_F (Fermi Level) $f(E_F) = 0.5$ $T = 0$ K (Step Function) $T > 0$ K (Thermal Smearing $\sim k_B T$)

FIGURE 4.2: STEP FUNCTION AT $T = 0$ K AND THERMAL BROADENING AT $T > 0$ K WITH INVARIANT POINT $F(E_F) = 0.5$

5. Thermionic Emission: The Richardson-Dushman Equation

Physical Mechanism & Richardson-Dushman Equation Derivation

Thermionic Emission: The thermal ejection of conduction electrons from a heated metal surface when thermal energy supplied exceeds the metal's **Work Function ϕ** ($\phi = E_{\text{vacuum}} - E_F$).

To escape across the metal surface barrier perpendicular to the x-axis, an electron must possess kinetic energy: $1/2 m v_x^2 \geq E_F + \phi$.

The total emission current density J is obtained by integrating over all velocities obeying this barrier condition using Fermi-Dirac statistics:

$$J = e \int_{v_{x,\min}}^{\infty} v_x \cdot n(v_x) dv_x = \left[4 \pi m e k_B^2 / h^3 \right] \cdot T^2 \cdot \exp[- \phi / (k_B T)]$$

$$\text{Richardson-Dushman Equation: } J = A \cdot T^2 \cdot e^{- \phi / (k_B T)}$$

- **$A = 4\pi m e k_B^2 / h^3$** : Universal Richardson constant ($\approx 1.2 \times 10^6 \text{ A}\cdot\text{m}^{-2}\cdot\text{K}^{-2} = 120 \text{ A}\cdot\text{cm}^{-2}\cdot\text{K}^{-2}$).
- **ϕ** : Metal work function (typically 2 to 5 eV; e.g., Tungsten $\phi \approx 4.54$ eV, Thoriated Tungsten $\phi \approx 2.6$ eV).
- **T** : Absolute temperature of cathode in Kelvin.

Key Practical Insight: Current density increases exponentially with temperature and decreases exponentially with work function, which explains why cathodes in electron microscopes and vacuum tubes are coated with low-work-function alkaline oxides (BaO, SrO).